



80%

FIRST CLASS BENG

91%

VAIDAR CAPSTONE

IET

PRIZE WINNER 2026

Chess

HARDWARE TEST

Accomplished **Hardware Test Engineer & FPGA Verification Specialist** with a First-Class 80% BEng (Hons) in Electronic and Computer Engineering and recipient of **The IET Prize 2026**. Technical expertise spanning military-grade electro-optical surveillance director qualification, automated hardware-in-the-loop (HIL) testbench design, and automated digital logic verification. Developer of **VAIDAR**, a novel Python-native execution engine bridging machine learning models with physical AMD Xilinx Artix-7 FPGA silicon for accelerated coverage closure. Highly experienced in full-lifecycle Integration, Verification & Validation (IV&V), IBM DOORS requirement traceability, and multidisciplinary defect triage.

TECHNICAL COMPETENCIES MATRIX

Core Engineering Tooling

FPGA & Digital Logic

Silicon

Verilog HDL, IEEE VHDL, AMD Xilinx Vivado, Intel Quartus, Artix-7, Zynq-7000, ModelSim RTL, Timing Closure (XDC Constraints), STA, FSM Architecture.

HIL & Test Automation

Automation

Python HIL Test Engines, PySerial, Automated Test Equipment (ATE), FAT/SAT/ATP Protocols, Signal Injection Rigs, Real-Time Telemetry & Log Triage.

Electro-Optics & Systems

Optronics

Hawkeye Surveillance Directors, Daylight HD Optics, Cooled Thermal Imagers, Eye-Safe Laser Rangefinders, Optical Collimation, ESS Vibration & Thermal.

Protocols & Instruments

Hardware

Digital Storage Oscilloscopes, Logic Analysers, Spectrum Analysers, JTAG Boundary Scan, RS-422, CAN Bus, Ethernet, SPI, I2C, UART, PCIe, Camera Link.

PROFESSIONAL WORK EXPERIENCE

Defence Industry

Hardware Test Engineer

Current Role

June 2026 – Present

Chess Dynamics • Horsham, West Sussex, UK

Full-Time Industry

Specialist defence and security enterprise manufacturing high-precision electro-optical surveillance, tracking directors, and stabilised gimbals for international allied defence programmes.

Hardware Qualification & Test Execution:

Coordinate and execute comprehensive hardware test activities, advising project managers and systems engineering leads on test scoping, verification feasibility, and qualification schedules.

Factory Acceptance Testing (FAT):

Coordinate and execute formal Factory Acceptance Testing on Hawkeye electro-optical directors, verifying daylight HD optics, cooled continuous-zoom thermal cameras, and eye-safe laser rangefinders against military customer specifications.

Hardware Diagnostics & Fault Isolation:

Conduct board-level fault diagnostics and root-cause isolation across multi-layer PCBs, motor control servo feedback loops, slip-ring comms (RS-422, CAN, Ethernet), and high-bandwidth video processing pipelines.

Optical Boresighting & Calibration:

Perform precision optical boresighting and laser rangefinder alignment routines using optical collimators and blackbody emitters to achieve exact weapon-system integration.

Environmental Qualification (ESS):

Coordinate environmental stress screening, including climatic chamber thermal profiling (-40°C to +70°C) and multi-axis vibration testing to military standards.

Documentation & Procedures:

Author formal Factory Acceptance Test reports, engineering discrepancy logs, and Standard Operating Procedures (SOPs).

Hardware Test Engineering

Factory Acceptance Testing

Optical Boresighting

RS-422 / CAN / Ethernet

Thermal & Vibration ESS

Systems Engineer — IV&V Placement

12-Month Placement

June 2024 – June 2025

BAE Systems • Rochester, Kent, UK

Global tier-one defence and aerospace enterprise delivering mission-critical electronic systems and active vehicle electronics.

- ▶ **IV&V Lifecycle Execution:** Executed end-to-end Integration, Verification & Validation (IV&V) procedures on safety-critical electronic systems hardware and testbench architectures.
- ▶ **Python Test Automation:** Developed modular Python-based automated test scripts for lab testbenches and signal injection hardware, significantly accelerating regression testing throughput.
- ▶ **Requirements Traceability (IBM DOORS):** Managed bidirectional requirements verification matrices in IBM DOORS, mapping high-level system requirements to granular hardware test cases with 100% audit traceability.
- ▶ **Signal Integrity & Simulation:** Verified electronic hardware subsystems using signal injection, high-bandwidth oscilloscopes, and MATLAB/Simulink models, including Frequency Response Analysis (FRA).
- ▶ **Anomaly Debugging & ERB Reviews:** Collaborated with multidisciplinary hardware, software, and systems teams to diagnose hardware root-cause anomalies, authoring defect reports presented at Engineering Review Boards.
- ▶ **STEM Ambassador:** Delivered interactive "STEM in a Box" engineering workshops to over 400 school students across Kent as a certified STEM Ambassador.

Python Automation

IBM DOORS

IV&V Lifecycle

Frequency Response Analysis

Signal Injection

VAIDAR: AI-Driven Hardware-in-the-Loop FPGA Verification Engine

Flagship Project

AMD Xilinx Artix-7

Final Year Capstone Project • First Class 91% (A+) • The IET Prize Winner 2026

Architected an automated hardware-in-the-loop verification engine bridging host software directly to physical FPGA silicon over high-speed 921,600-baud UART to eliminate the 70% lifecycle bottleneck of software simulation.

Physical Hardware Testbed

Real-time vector streaming between host Python runtime and synthesised 16-bit ALU on physical Digilent Nexys A7 FPGA silicon.

Machine Learning Stimulus

Integrated adaptive stimulus generation and automated LLM discrepancy triage, delivering 99.9%+ logic coverage closure.

▶ Achieved **968.9 tests/second** sustained throughput on physical silicon (~44,000x faster than manual bench testing with 100% physical accuracy).

EDUCATION

BEng (Hons) Electronic & Computer Engineering
University of Brighton • First Class 80%
Graduated July 2026
Level 6: 80% • Level 5: 77% • Level 4: 78%
Specialisms: Digital RTL, Embedded C, HIL Automation, DSP, Closed-Loop Control.

OFFICIAL HONOURS

The IET Prize 2026
Awarded by The Institution of Engineering and Technology for outstanding academic distinction (First Class 80%).

IET Sussex Prize 2023
Awarded by The IET for outstanding first-year academic performance.

COMMERCIAL LEADERSHIP & OPERATIONS

Brighton Real Ale Society (Oct 2025 – June 2026)
Co-President, Vice President & Web Architect
Elected executive leader managing society administration, event licensing, budgeting, and engineered the full-stack web platform using Next.js, Prisma, and Supabase.

bp / M&S Simply Food (Dec 2023 – June 2026)
Customer Service Assistant
Forecourt safety compliance, hazardous materials handling, fresh food temperature audits, and inventory auditing.

McDonald's (Aug 2020 – Mar 2024)
Shift Manager — Eastleigh, Hampshire
Led operations directing shifts of up to 30 personnel under intense tempo, shift scheduling, safe drops, cash reconciliation, and hygiene auditing.

Deliveroo & Jamie's Computers
Urban Courier & PC Refurbisher
Time-critical urban food delivery (Sept 2023 – June 2024); diagnostic testing and hardware refurbishing for low-income families (2021).